

Pre-Registration: EFC $dN/dM(z>7)$ Halo Mass Function via $\mu(a)$ Perturbation Channel

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Validation candidate: `vc_de9a6ee1deb6`

Abstract. This document is a sealed pre-registration of an Energy-Flow Cosmology (EFC) prediction for the high-redshift halo mass function $n(M, z)$ at $z = 7\text{--}12$, derived through the EFC $\mu(a)$ perturbation channel established in EFCLASS Technical Notes I and II. The prediction operates in the L3 regime per the EFC Atlas (RCMP classification of `early_structure_JWST`), where the S_{flow} -boost mechanism is the canonical EFC role for high- z structure formation. EFC modifies the linear growth equation via an effective gravitational coupling $\mu(a) = 1 - B g(a)$ acting on the Poisson source, where $g(a)$ is the EFC gate function calibrated against background and perturbation data under Entropy-Bounded Empiricism (EBE) conditions. The HMF prediction follows from an explicit five-step derivation chain $g(a) \rightarrow \mu(a) \rightarrow D(a) \rightarrow \sigma(M, z) \rightarrow n(M, z)$, ensuring the result is structurally derived rather than parameter-fitted at the HMF level. We pre-register the full chain, the falsification criteria ($\Delta\chi^2 < 4$ acceptance, $>3\sigma$ disagreement falsification against JWST CEERS / COSMOS-Web / GLASS counts), three independent halo-mass-function prescriptions (Press-Schechter, Sheth-Tormen, Tinker 2008/2010) for HMF-modeling robustness, the datasets to be confronted, and the sealing protocol. Numerical curves at $z = 7, 8, 9, 10, 11, 12$ are computed once (B, z_t, n) are fixed by the joint $S(z)$ -fit (validation candidate `vc_4ea464e38b5f`) and sealed by SHA-256 hash before any data fitting begins. This pre-registration closes a gap in the EFC corpus where the JWST early-structure phenomenon was previously addressed only observationally (Magnusson 2026 COSMOS, DOI [31059964](https://doi.org/31059964)) without a formal quantitative prediction. Full derivation paper to follow.

Keywords: energy-flow cosmology, halo mass function, JWST, high-redshift structure, $\mu(a)$ perturbation channel, RCMP-L3, EBE, sealed prediction, pre-registration

1 Pre-registration scope and what is sealed

What this document is. A pre-registered sealed prediction in the sense established by EFC's prior pre-registration DOIs ([32023788](https://doi.org/32023788) $\text{SO} \times \text{Euclid EG}$; [32013738](https://doi.org/32013738) Rubin DP2 shear; [32013156](https://doi.org/32013156) DESI $f\sigma_8$). The deliverable is a structurally derived prediction with explicit pass/fail criteria, sealed before any confrontation with the JWST counts data.

What this document is not. A full derivation paper. The $\mu(a)$ perturbation channel is established in Technical Note II (DOI [31333600](https://doi.org/31333600)) and the gate function calibration in Technical Note I (DOI [31333414](https://doi.org/31333414)); the present document chains these to the HMF. The numerical $n(M, z)$ curves are computed once the gate parameters (B, z_t, n) are fixed by the joint $S(z)$ -fit (validation candidate `vc_4ea464e38b5f`, currently pending); that step is not yet executed and is not sealed in this document. What is sealed: the chain, the dataset selection, the falsification criteria, and the parametric form into which the gate parameters enter.

Why pre-register before computing the curves. The structure of the prediction — and crucially, the structural chain that produces it — can be locked now. Once (B, z_t, n) are determined from the joint $S(z)$ -fit and the curves are computed, the result is over-determined: any post-hoc adjustment would break the chain published here. This is the same protocol used

for the EG, Rubin, and DESI pre-registrations.

Regime placement. Per the EFC Atlas, the phenomenon `early_structure_JWST` is an L3-regime observable (high- z mass function), addressed by EFC via the `S_flow_boost_at_high_z` mechanism. This pre-registration tests the L3 regime under L3-appropriate formalism: the $\mu(a)$ perturbation channel reaches its L3-asymptotic form for $z \gg z_t$ with transition sharpness controlled by the exponent n . Both z_t and n are joint-fit parameters; the sensitivity of the HMF prediction to these parameters falls out of the sealed numerical table once the joint-fit posterior is determined, and is reported alongside the table in the SHA-256 addendum. Confrontation with data is regime-internal, not cross-regime extrapolation.

Scope: cosmology channel only. JWST observations require a stellar-mass-to-halo-mass mapping that involves baryon physics, feedback, dust, and IMF assumptions outside this pre-registration’s scope. We compare EFC and Λ CDM predictions *under identical mapping assumptions*, isolating the cosmology channel. A failure within this scope is a failure of the cosmology channel as predicted by EFC; it does not, by itself, falsify EFC’s deeper claims that depend on alternative mapping conventions. This convention follows standard modified-gravity practice (Bellini-Sawicki, Horndeski).

2 The EFC perturbation channel and prediction chain

2.1 Background-level gate function

EFCLASS Technical Note I (DOI [31333414](#)) establishes the background-level modification:

$$\Delta E^2(a) = A[g(a) - g(0)], \quad (1)$$

acting in the Friedmann equation. The sign lemma proves $\Delta E^2 \leq 0$ for all $z > 0$ when g is monotonically increasing in a . The gate function is parameterised as

$$g(a) = \frac{1}{1 + (z_t/z(a))^n}, \quad z(a) = 1/a - 1, \quad (2)$$

with transition redshift z_t and steepness exponent n .

2.2 Perturbation channel

EFCLASS Technical Note II (DOI [31333600](#)) establishes the EFC perturbation modification as an effective gravitational coupling on the Poisson source:

$$\mu(a) = 1 - B g(a), \quad (3)$$

with $\mu(a) < 1$ generically (suppressed structure formation in the gate-active regime). The amplitude B is the perturbation-channel parameter; together with (z_t, n) from the background gate, this gives a three-parameter system (B, z_t, n) .

2.3 Modified linear growth equation

The linear growth rate $f = d \ln D / d \ln a$ obeys the modified ODE (implemented in the EFC code module `src/efc/perturbation/growth.py`):

$$\boxed{f' + f^2 + \left(\frac{1}{2} - \frac{3}{2}\tilde{\Omega}_m\right)f = \frac{3}{2}\mu(a)\tilde{\Omega}_m} \quad (4)$$

where $\tilde{\Omega}_m(a) = \Omega_m a^{-3}/E^2(a)$ and $\mu(a)$ is given by Eq. (3). The growth factor $D(a)$ is recovered by integration of Eq. (4) from a deep-matter-era initial condition, with the GR limit $\mu \rightarrow 1$ recovering Λ CDM growth.

2.4 The HMF chain

The full prediction chain is:

$$g(a; z_t, n) \longrightarrow \mu(a; B, z_t, n) \longrightarrow D(a) \longrightarrow \sigma(M, z) \longrightarrow n(M, z) \quad (5)$$

1. **Gate** $\rightarrow \mu$. Eq. (3) with parameters from the joint $S(z)$ -fit.
2. $\mu \rightarrow D$. Integration of the modified growth ODE (4), no additional parameters.
3. $D \rightarrow \sigma$. Standard variance integral

$$\sigma^2(M, z) = \int \frac{d^3k}{(2\pi)^3} P_{\text{lin}}(k, z=0) |W(kR(M))|^2 D^2(z), \quad (6)$$

with $D(z)$ from step 2.

4. $\sigma \rightarrow n$. Press-Schechter, Sheth-Tormen, and Tinker (2008/2010) prescriptions:

$$n(M, z) dM = \frac{\bar{\rho}}{M} f(\nu) \frac{d\nu}{dM} dM, \quad \nu \equiv \delta_c / \sigma(M, z). \quad (7)$$

The three prescriptions are computed in parallel from the same $\sigma(M, z)$; differences across them quantify HMF-modeling uncertainty.

The chain is structurally rigid: once (B, z_t, n) are fixed by the joint fit, every downstream quantity is determined. There is no fitting freedom at the HMF level. The sole free parameter is the choice of reference linear power spectrum $P_{\text{lin}}(k, z=0)$, which is fixed to the same Planck-anchored cosmology used in the joint fit.

Initial conditions. We anchor the linear power spectrum on Planck-recombination initial conditions, which are the best-measured CMB-era observables and lie outside the scope of EFC modification. EFC's $\mu(a)$ acts on subsequent linear growth, not the primordial spectrum. This is standard modified-gravity practice and ensures that the EFC vs Λ CDM comparison is driven by the growth channel alone.

EBE conditions. Parameter determination via the joint $S(z)$ -fit operates under Entropy-Bounded Empiricism (EBE) conditions, the EFC calibration discipline documented in the EBE-SPARC175 corpus. EBE prevents over-fitting by bounding parameter excursions to entropy-consistent regions of the joint constraint space. The growth-sector closure (DOI [32084670](https://doi.org/10.26434/chemrxiv-2020-08-01)) is shown to hold under EBE conditions; the present pre-registration inherits this discipline.

2.5 Numerical predictions to be sealed

Upon fixing (B, z_t, n) from the joint $S(z)$ -fit, this pre-registration commits to producing $n(>M_*, z)$ tables at the following grid:

Redshifts:	$z \in \{7, 8, 9, 10, 11, 12\}$
Mass thresholds:	$\log_{10}(M_*/M_\odot) \in \{8.5, 9.0, 9.5, 10.0, 10.5, 11.0\}$
Three prescriptions:	Press-Schechter, Sheth-Tormen, and Tinker (2008/2010)

The full $6 \times 6 \times 3 = 108$ -entry table is the sealed prediction object. Its SHA-256 hash will be appended to this document as Appendix A in the deposited version, before any comparison to JWST data.

Why three prescriptions. Press-Schechter and Sheth-Tormen are analytic and primarily calibrated on simulations at lower redshifts. Tinker (2008/2010) is calibrated against high-resolution N-body simulations spanning higher redshifts and provides a more reliable HMF

normalisation in the JWST regime. Including all three makes the test robust against HMF-modeling choice: a pass requires consistency across at least two of the three prescriptions; a failure is unambiguous only when all three disagree with data.

HMF-prescriptions inherit GR-collapse assumptions. All three prescriptions are calibrated against N-body simulations under General Relativity collapse dynamics. EFC’s $\mu(a)$ modification of the Poisson source is small in absolute terms ($|1 - \mu(a)| \ll 1$ in the regime tested by Tech Note II) and the linear collapse threshold δ_c is therefore expected to shift by at most the few-percent level documented in modified-gravity HMF studies (Schmidt 2009 and follow-ups). The HMF prescriptions retain their universal-form behavior; the dominant EFC effect propagates through $D(a)$ to $\sigma(M, z)$ and is captured by the chain in Eq. (5). We treat collapse-physics modification as a known modeling-systematic that motivates the three-prescription robustness test rather than as a falsifying ambiguity. A formal MG-collapse correction to δ_c is identified as an open theoretical task and listed in the companion derivation paper.

3 Datasets to be confronted

- **Labé et al. 2023**, *Nature* — COSMOS-Web stellar mass density at $z = 7$ –10.
- **Harikane et al. 2024**, *ApJ* — JWST UV luminosity functions to $z \sim 14$.
- **CEERS-2024 public catalog** — spectroscopically confirmed $z > 7$ galaxies.
- **Boylan-Kolchin 2023**, *Nature Astronomy* — mass-budget tension framework providing Λ CDM baseline.
- **Magnusson 2026 COSMOS** (DOI [31059964](https://doi.org/10.6084/m9.figshare.32149654)) — halo-limited cap argument and observational anchor for the EFC interpretation.

4 Pass/fail criteria

Statistical definitions. The χ^2 statistic is computed as $\chi^2 = \mathbf{\Delta}^T \mathbf{C}^{-1} \mathbf{\Delta}$, where $\mathbf{\Delta}_i = n_{\text{obs}}(>M_*, z_i) - n_{\text{EFC}}(>M_*, z_i)$ over N data points and $\mathbf{C}$ is the data covariance matrix including catalog statistical uncertainty and Poisson contribution per redshift bin. Number of free parameters at the HMF level is $k = 0$ (since (B, z_t, n) are fixed by the upstream joint $S(z)$ -fit), giving degrees of freedom $\text{DOF} = N$. The exact data-vector size N depends on the catalog versions current at step 6 of the sealing protocol and is reported alongside the SHA-256-sealed prediction table.

Acceptance. The EFC HMF prediction is accepted with $\Delta\chi^2 < 4$ against the joint dataset listed above (where $\Delta\chi^2 \equiv \chi_{\text{EFC}}^2 - \chi_{\Lambda\text{CDM}}^2$), with (B, z_t, n) fixed by the joint $S(z)$ -fit and *not* re-fitted at the HMF level. Acceptance requires consistency under at least two of the three HMF prescriptions (Press-Schechter, Sheth-Tormen, Tinker).

Falsification. The EFC HMF prediction is falsified by $>3\sigma$ disagreement across the z -grid (combined χ^2 test, evaluated against $\text{DOF} = N$) under all three prescriptions, or by inconsistency requiring any of (B, z_t, n) to differ from the joint-fit values by more than its 1σ uncertainty. Single-prescription disagreement is not falsification: it indicates HMF-model dependence, which is treated separately.

Geometry-locking constraint. The growth-sector closure (DOI [32084670](https://doi.org/10.6084/m9.figshare.32149654)) demonstrates that once geometry is locked by BAO + cosmic-chronometer $H(z)$, the EFC growth-coupling structure is constrained: the three coupling classes (multiplicative G_{eff} , additive local friction, additive non-local memory friction) are either structurally excluded or absorbed by the Ω_m -coupling degeneracy. Consequently, the $\mu(a)$ -channel parameters (B, z_t, n) are constrained — not

eliminated — in the post-geometry-fixing space. HMF-level disagreement under fixed mapping and at least two prescriptions is therefore a genuine theory failure within the cosmology channel, not a parameter-shuffle escape.

Structural robustness. The robustness analysis (DOI [31332730](https://doi.org/10.6084/m9.figshare.32149654)) confirms the EFC growth-sector signal survives leave-one-out, sound-horizon-independent, and amplitude-prior-controlled tests at $\geq 1.84\sigma$ across all variants. The $\mu(a)$ -channel formalism used in this pre-registration inherits this robustness.

Independent growth-channel cross-check. The same $\mu(a)$ mechanism that drives the HMF prediction here is independently tested via the growth rate $f\sigma_8(z)$ in the DESI DR2 full-shape RSD pre-registration (DOI [32013156](https://doi.org/10.6084/m9.figshare.32013156), sealed prediction $f\sigma_8(z = 0.7) = 0.430$). That test isolates the growth channel without invoking HMF prescriptions or stellar-to-halo mapping. Joint pass on both pre-registrations would constitute a double-channel confirmation of the same $\mu(a)$ modification; joint failure would localize the problem to the shared mechanism rather than to HMF-modeling or mapping systematics.

5 Sealing protocol

1. This document is deposited on figshare with DOI [10.6084/m9.figshare.32149654](https://doi.org/10.6084/m9.figshare.32149654) before the joint $S(z)$ -fit is executed.
2. The companion derivation paper (gate $g(a)$ and $\mu(a)$ specific to the high- z regime, with reduction to Tech Note II form) is completed.
3. The joint $S(z)$ -fit (validation candidate [vc_4ea464e38b5f](https://doi.org/10.6084/m9.figshare.32149654)) determines (B, z_t, n) .
4. The 108-entry $n(>M_*, z)$ table (six redshifts $\times$ six masses $\times$ three prescriptions) is computed via the chain in Eq. (5).
5. The table is hashed (SHA-256) and the hash is published as an addendum DOI to this pre-registration.
6. Only after step 5 is the prediction confronted with JWST data.

Steps 1–5 are pre-data-confrontation. Step 6 is the actual test. Any modification to the prediction between steps 5 and 6 invalidates the seal.

Dataset-revision caveat. JWST stellar-mass and redshift catalogs are subject to ongoing revision (improved photo- z calibration, alternative SED-fitting, selection-completeness corrections). We commit to using the catalog versions current at the time of step 6 confrontation; substantive subsequent catalog revisions trigger an addendum DOI documenting which version was used and why, without re-fitting any of (B, z_t, n) .

6 Relation to the EFC validation ledger

Closes. Validation candidate [vc_de9a6ee1deb6](https://doi.org/10.6084/m9.figshare.32149654) (status: PIPELINED, awaiting Paper 1 DOI). Closes the registered gap “Early structure JWST — quantitative prediction paper” in the EFC Gap Analysis. Companion to the joint $S(z)$ -fit validation [vc_4ea464e38b5f](https://doi.org/10.6084/m9.figshare.32149654).

Supersedes. Two earlier candidate formulations: [vc_eea84fbbf592](https://doi.org/10.6084/m9.figshare.32149654) (rejected — single-observable test, subsumed under the cross-observable joint $S(z)$ -fit) and [vc_3937e965d27a](https://doi.org/10.6084/m9.figshare.32149654) (rejected — missing explicit growth chain, replaced by the present formulation with all five chain steps explicit and grounded in EFCLASS Technical Notes).

Anchored to. The joint $S(z)$ -fit candidate [vc_4ea464e38b5f](https://doi.org/10.6084/m9.figshare.32149654) (“ $S(z)$ single-parameter joint fit: CMB anomaly $\leftrightarrow$ JWST early-galaxy excess”), which determines (B, z_t, n) . The CMB- $T(z)$

scaling test [vc_7299eae1d6b](#) provides an independent constraint on the same gate function from the SZ-cluster channel. The growth-sector closure (DOI [32084670](#)) provides the geometry-locking proof that prevents post-hoc tuning.

Companion to. The Unified Linear Perturbation Theory pre-registration (validation candidate [vc_3c5f8548af90](#)), which closes the kSZ velocity gap and the $H(z)$ Level-3 background recovery gap from the same FRW linearization. The two pre-registrations together close three of the eight HIGH-priority gaps in the April 2026 gap analysis.

7 Timeline

The schedule below is anchored to the deposit date of this pre-registration (2026 Q2). Calendar quarters are used rather than relative weeks so that the timeline remains meaningful when the document is read at any future date.

- **2026 Q2 (May–Jun):** Pre-registration deposit. DOI assigned by figshare.
- **2026 Q3 (Jul–Sep):** Companion derivation paper showing the high- z regime reduction of the $\mu(a)$ -channel; joint $S(z)$ -fit pipeline executed; (B, z_t, n) determined.
- **2026 Q4 (Oct–Dec):** Numerical $n(M, z)$ table computed via Eq. (5) and sealed by SHA-256 addendum DOI.
- **2027 Q1 (Jan–Mar):** Confrontation with JWST data; pass/fail outcome documented in EFC Validation Ledger and reflected in the $n(M, z)$ table publication.

Substantive deviations from this schedule are documented as addendum entries on the present DOI. The schedule does not affect the seal: the sealing protocol guarantees that no parameter or prediction is adjusted between sealing (step 5) and data confrontation (step 6), regardless of when the latter occurs.

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Pre-registration document v4 (revised after two rounds of pre-deposit peer-style review). Sealed parametric form using the $\mu(a)$ perturbation channel established in EFCLASS Technical Notes I & II, anchored to RCMP-L3 Atlas regime, calibrated under EBE conditions, tested across three HMF prescriptions with explicit MG-collapse caveat, and cross-linked to the independent $f\sigma_8(z)$ growth-channel pre-registration (DOI 32013156). Numerical $n(M, z)$ table to follow as SHA-256 addendum once (B, z_t, n) are determined by the joint $S(z)$ -fit `vc_4ea464e38b5f`. Living entry: EFC Validation Ledger `vc_de9a6ee1deb6`.